Proof by Induction: Matrices

Questions 1–15

Question 1. Show that

a) $\begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 2k+1 & -4k \\ k & 1-2k \end{pmatrix} = \begin{pmatrix} 2(k+1)+1 & -4(k+1) \\ k+1 & 1-2(k+1) \end{pmatrix}$

b) $\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} 1 & 2^k-1 \\ 0 & 2^k \end{pmatrix} = \begin{pmatrix} 1 & 2^{k+1}-1 \\ 0 & 2^{k+1} \end{pmatrix}$

c) $\begin{pmatrix} 1 & 0 \\ -1 & 5 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ -\frac{1}{4}(5^k-1) & 5^k \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -\frac{1}{4}(5^{k+1}-1) & 5^{k+1} \end{pmatrix}$

Question 2. Prove by induction, for all integers $n \geq 1$, that $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^n = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$.

Question 3. Given $A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$, prove for all integers $n \geq 1$ that $A^n = \begin{pmatrix} 1 & 2n \\ 0 & 1 \end{pmatrix}$.

Question 4. Prove by mathematical induction, for $n \in \mathbb{N}$, that $\begin{pmatrix} 3 & -2 \\ 2 & -1 \end{pmatrix}^n = \begin{pmatrix} 2n+1 & -2n \\ 2n & 1-2n \end{pmatrix}$.

Question 5. Prove by mathematical induction, for $n \in \mathbb{N}$, that $\begin{pmatrix} 3 & 2i \\ 2i & -1 \end{pmatrix}^n = \begin{pmatrix} 2n+1 & 2ni \\ 2ni & 1-2n \end{pmatrix}$.

Question 6. Prove by mathematical induction, for $n \in \mathbb{N}$, that $\begin{pmatrix} 2 & 2 \\ 0 & 4 \end{pmatrix}^n = \begin{pmatrix} 2^n & 4^n - 2^n \\ 0 & 4^n \end{pmatrix}$.

Question 7. Given that $A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$, prove for $n \in \mathbb{N}$ that $A^n = \begin{pmatrix} 2^n & n2^{n-1} \\ 0 & 2^n \end{pmatrix}$.

Question 8. Given that $A = \begin{pmatrix} 1 & 1 \\ 0 & 3 \end{pmatrix}$, prove for $n \in \mathbb{N}$ that $A^n = \begin{pmatrix} 1 & \frac{1}{2}(3^n - 1) \\ 0 & 3^n \end{pmatrix}$.

Question 9. Prove by mathematical induction, for $n \in \mathbb{N}$, that $\begin{pmatrix} 1 & 0 \\ -1 & 4 \end{pmatrix}^n = \frac{1}{3}\begin{pmatrix} 1 & 0 \\ 1-4^n & 4^n \end{pmatrix}$.

Question 10. Given that $A = \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$,

a) Find A^2, A^3 and A^4.
b) Write down a suitable form for A^n where n is a positive integer.
c) Prove your answer to part b).

Question 11. Given that $A = \begin{pmatrix} 4 & a \\ 0 & 1 \end{pmatrix}$ where a is real, prove $A^n = \begin{pmatrix} 4^n & \frac{a}{3}(4^n - 1) \\ 0 & 1 \end{pmatrix}$ is true for all positive integers n.

Question 12. Given that $A = \begin{pmatrix} -2 & -9 \\ 1 & 4 \end{pmatrix}$,

a) Prove that for $n \in \mathbb{N}$, $A^n = \begin{pmatrix} 1 - 3n & -9n \\ n & 3n + 1 \end{pmatrix}$.

b) Find the line of invariant points of the transformation represented by matrix A.

c) Show, algebraically, that the transformation A^n has the same line of invariant points.

Question 13. It is known that $A = \begin{pmatrix} 2 & -a \\ a & 2 \end{pmatrix}$ where a is imaginary and $\det A = 0$.

a) Find the value of a.

b) Prove by mathematical induction, for $n \in \mathbb{N}$, $A^n = 4^{n-1}A$.

Question 14. Two square matrices, A and B, are such that $AB = B^4A$ and $B^5 = I$. Prove that $B^nAB^n = A$ for $n \in \mathbb{N}$.

Question 15. Prove by mathematical induction, for $n \in \mathbb{N}$, that

a) $\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}^n = \begin{pmatrix} 1 & n & \frac{1}{2}n(n+1) \\ 0 & 1 & n \\ 0 & 0 & 1 \end{pmatrix}$

b) $\begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}^n = \begin{pmatrix} 1 & 2n & \frac{1}{2}n(n+3) \\ 0 & 1 & n \\ 0 & 0 & 1 \end{pmatrix}$

c) $\begin{pmatrix} 2 & -1 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}^n = \begin{pmatrix} 2^n & 1-2^n & 3(2^n-1) \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

Proof by Induction: Divisibility

Questions 1–17

Question 1. Show that each of the following are divisible by 4.

a) $(2n+1)(2n-3)-4n+23$

b) $16^{3k}-4$

c) $2^{k+2}-4\times 3^k+8$

Question 2. Show that each of the following are divisible by 6.

a) $6^{k+1}-18$

b) $2^{k+1}\times 3^{k+2}$

c) $(k^2+1)(k-4)+10(k^2+k+1)$

Question 3. Given that a, b and c are positive integers, decide whether each of the following statements are true or false:

a) If a and b are factors of c then $a + b$ is a factor of c.
b) If a is a factor of both b and c then a is a factor of $b + c$.
c) If a is a factor of $b + c$ then both b and c are factors of a.
d) If a is a factor of b and b is a factor of c then a is a factor of $b + c$.

Question 4. Prove, by induction, that $n^2 + n$ is divisible by 2 for all integers $n \geq 1$.

Question 5. Using proof by induction, prove that

a) $n^3 - n$ is divisible by 3

b) $n^3 + 3n^2 + 2n$ is divisible by 6 for all integers $n \geq 1$

Question 6. Prove, by induction, that $9^n + 7$ is divisible by 8 for all integers $n \geq 1$.

Question 7. Using proof by induction, prove that $5^n - 4n + 3$ is divisible by 4 for all integers $n \geq 0$.

Question 8. Using proof by induction, prove that $5^n + 4n - 1$ is divisible by 8 for all integers $n \geq 1$.

Question 9. Prove, by induction, that $6^n - 2^n$ is a multiple of 4 for all integers $n \geq 1$.

Question 10. Prove, by induction, that

a) $5^n + 3 \times 9^n$ is divisible by 4

b) $5^{2n} + 3^{2n-1}$ is divisible by 4 for all integers $n \geq 1$

Question 11. Prove, by induction, that $15^n - 8^{n-2}$ is divisible by 7 for all integers $n \geq 2$.

Question 12. Prove, using induction, that for all positive odd integers n, $3^n + 7^n$ is divisible by 10.

Question 13. The following shows a student's answer to the question "Prove, by induction, that $3^{2n}-1$ is a multiple of 8 for all integers $n \geq 1$".

True for $n = 1$
When $n = k + 1$
$(3^{2(k+1)} - 1) - (3^{2k} - 1)$
$= 3^{2k} \times 3^2 - 3^{2k} = 3^{2k}(9 - 1) = 8 \times 3^{2k}$
which is a multiple of 8 so true for all $n \geq 1$

a) Find all the errors.
b) Correct the proof.

Question 14. Prove by induction that the sum of the cubes of three consecutive positive integers is divisible by 3.

Question 15. For $n \geq 1$, it is given that $f(n) = 50n^2 + 70n$.

a) Show that $f(n+1) - f(n) = 20(5n+6)$.

b) Hence prove by induction that $f(n)$ is divisible by 4 for all integers $n \geq 1$.

Question 16. For $n \geq 1$, it is given that $f(n) = 24 \times 2^{4n} + 3^{4n}$.

a) Find $f(n+1) - f(n)$.

b) Hence prove by induction that $f(n)$ is divisible by 5 for all integers $n \geq 1$.

Question 17.

a) Prove that for all integers $n \geq 1$, $f(n) = 2n^3 - 5n$ is a multiple of 3.

b) Show that $f(-x) = -f(x)$ for all real values of x.

c) Hence prove that $f(n)$ is a multiple of 3 for all integer values of n.

Proof by Induction: Sum of Series

Questions 1–15

Question 1. Show that

a) $\frac{1}{3}k(k+1)(k+2) + (k+1)(k+2) = \frac{1}{3}(k+1)(k+2)(k+3)$

b) $\frac{k}{k+1} + \frac{1}{(k+1)(k+2)} = \frac{k+1}{k+2}$

c) $(k+1)! - 1 + (k+1)(k+1)! = (k+2)! - 1$

Question 2. Prove by mathematical induction, for $n \in \mathbb{N}$:

a) $\sum_{r=1}^{n} 2^{r-1} = 2^n - 1$

b) $\sum_{r=1}^{n} 4^r = \frac{4}{3}(4^n - 1)$

Question 3. Prove by mathematical induction, for $n \in \mathbb{N}$:

a) $\sum_{r=1}^{n} r = \frac{1}{2}n(n+1)$

b) $\sum_{r=1}^{n} r^2 = \frac{1}{6}n(n+1)(2n+1)$

c) $\sum_{r=1}^{n} r^3 = \frac{1}{4}n^2(n+1)^2$

Question 4. Prove by mathematical induction, for $n \in \mathbb{N}$, $\sum_{r=1}^{n} r(r-1) = \frac{1}{3}n(n-1)(n+1)$.

Question 5. Prove by mathematical induction, for $n \in \mathbb{N}$, $-3-4-3+\cdots+n(n-4)=\frac{1}{6}n(n+1)(2n-11)$.

Question 6. Michael is partway through a proof, he writes: $\sum_{r=1}^{k+1} r(r+3)=\sum_{r=1}^{k} r(r+3)+(k+1)$. Identify his mistake and correct this line of working.

Question 7. Prove by mathematical induction, for $n \in \mathbb{N}$, $\sum_{r=1}^{n} \frac{1}{(r+1)(r+2)} = \frac{n}{2n+4}$.

Question 8. Prove by mathematical induction, for $n \in \mathbb{N}$, $\sum_{r=1}^{n} \frac{1}{(r+2)(r+4)} = \frac{n(7n+25)}{24(n+3)(n+4)}$.

Question 9. Prove by mathematical induction, for $n \in \mathbb{N}$:

a) $\sum_{r=1}^{n} \frac{1}{3^r} = \frac{1}{2} - \frac{1}{2 \times 3^n}$

b) $\sum_{r=1}^{n} \frac{r}{2^r} = 2 - \frac{n+2}{2^n}$

c) $\sum_{r=1}^{n} (r+1)2^r = n \times 2^{n+1}$

Question 10. It is conjectured that for constants a, b and c, $S_n = \sum_{r=1}^{n} (r+1)^2 = \frac{1}{6} n(an^2 + bn + c)$.

a) By considering S_1, S_2 and S_3 show that $a = 2$, $b = 9$ and $c = 13$.

b) Prove the conjecture by induction.

Question 11. Prove by mathematical induction, for $n \in \mathbb{N}$, $\sum_{r=1}^{n}(r+i) = -n + \frac{1}{2}n(n+1)i$.

Question 12. Prove by mathematical induction, for $n \in \mathbb{N}$, $\sum_{r=1}^{n} r \times r! = (n+1)! - 1$.

Question 13. Prove by mathematical induction, for $n \in \mathbb{N}$, $\sum_{r=1}^{n} \frac{r}{(r+1)!} = 1 - \frac{1}{(n+1)!}$.

Question 14. Prove, for $n \in \mathbb{N}$ and $a \in \mathbb{R}$, $\sum_{r=1}^{n} \frac{1}{(r+a)(r+a+1)} = \frac{n}{(1+a)(n+1+a)}$.

Question 15. Prove, for $n \in \mathbb{N}$ and $a \in \mathbb{R}$, $\sum_{r=1}^{n} a^r = \frac{a(a^n - 1)}{a - 1}$.